Rank 4 with torsion Z/2×Z/8

Problèmes ouverts — A. Dujella · 2026

Problem 2.2 (Dujella, 2026). Is there an elliptic curve over $\mathbb{Q}$ with torsion group $\mathbb{Z}/2\mathbb{Z}\times\mathbb{Z}/8\mathbb{Z}$ and rank equal to 4 ?

Remark. There are 34 known curves with torsion group $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/8\mathbb{Z}$ and rank equal to 3 : `https://web.math.pmf.unizg.hr/~duje/tors/z2z8.html`